MODULE 5

MIC

- Multivariable control
  - Concept
  - interactions and its effects
  - Modelling & transfer functions
  - Influence of interaction on possibility of feedback control
- Imp effects on Multivariable sys. behaviour
- Relative gain array
- Effect of interactn on stability
- Tuning of multi loop CS
- Model Based controllers
  - Internal model control
  - Model Predictive controller
  - Dynamic matrix controller
  - Self tuning controller.

MIMO

- Process inter^n happen in multiloop
- 1 MV മാറ്റിയാൽ multiple CV affect ആകും.
- to design, we need to determine interactions b/w loops.
  we need to pair MVs & CVs

for 2×2 system

2 i/ps, 2 o/ps

$$\frac{Y_1(s)}{U_1(s)} = G_{p11}(s), \quad \frac{Y_1(s)}{U_2(s)} = G_{p12}, \quad \frac{Y_2}{U_1} = G_{p21}, \quad \frac{Y_2}{U_2} = G_{p22}$$

$$Y_1(s) = G_{11}(s)\, U_1(s) + G_{12} U_2$$

$$Y_2 = G_{22} U_2 + G_{21} U_1$$

$$\begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} G_{p11} & G_{p12} \\ G_{p21} & G_{p22} \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \end{bmatrix}$$

$$Y(s) = G_p(s) \cdot U(s)$$

- If we have a 1-1/2-2 control scheme
  ↳ Y_1 controlled by U_1 &
  Y_2 — U_2.

then a disturbance changes Y_1, ~~causing U_1 to adjust~~ causing G_{c1} (controller in loop 1) to adjust U_1 to force Y_1 to SP. But this affects Y_2 via G_{p21}.

Since Y_2 changed, G_{c2} adjust U_2 to bring $Y_2 \to$ SP but causes change in Y_1 via G_{p12}. This causes entire s/m to come to new steady state.

~~init~~ initial change in ~~can you look the fuck in~~?

u_1 has 2 effects on Y_1.

1) a direct effect
2) an indirect effect via control loop interactions
   ↳ result of a 3rd, hidden feedback loop

- This can ~~cause~~
  - destabilize system
  - make controller tuning more difficult

INTERNAL MODEL CONTROL

"If controller has an exact model of process ~~past~~ then perfect control is theoretically achievable".

- IMC uses a process model to simulate process ~~and~~ response internally.
- compares real output from actual process with predicted output from mode.
- uses difference of the 2 to adjust control action.

IMC structure consists of:

- Process (P), Process model($\tilde{P}$), IMC controller (Q), feedback loop

20b) $A = \begin{bmatrix} \lambda_{11} & \lambda_{12} \\ \lambda_{21} & \lambda_{22} \end{bmatrix}$ $\quad y_1, y_2$

m_1, m_2

given $\lambda_{11} = 1$ we know ~~$\lambda_{12} = 1 - \lambda_{11} = 0$~~

since RGA is normalised, since sum of elements in row is 1,

~~$\lambda_{21} \Rightarrow \lambda$~~

$$\lambda_{11} + \lambda_{12} = 1 \Rightarrow \lambda_{12} = 0.$$

and sum of elements in a column is 1,

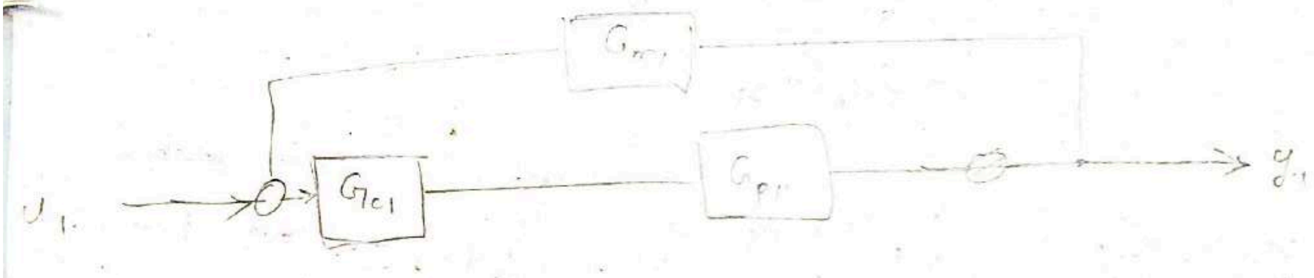
$$\lambda_{11} + \lambda_{21} = 1 \Rightarrow \lambda_{21} = 0.$$

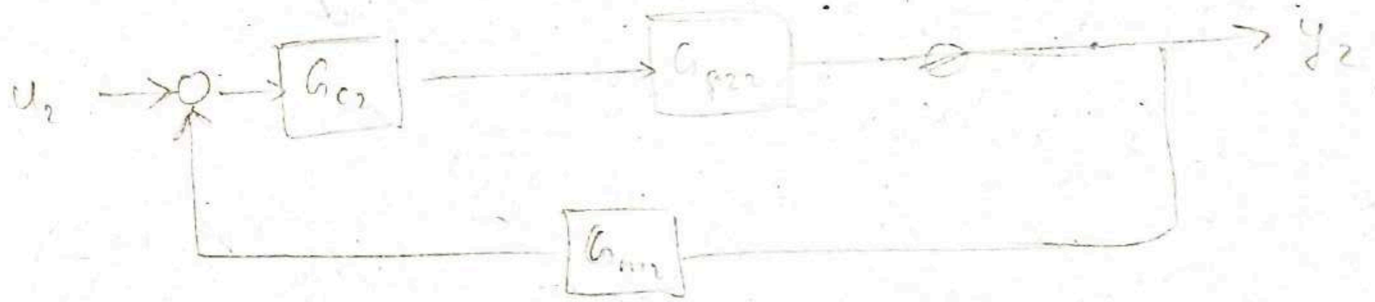
and $\lambda_{22} = 1 - 0 = \underline{1}$

$$\therefore A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Since $\lambda_{21} = 0 = \lambda_{12}$ there's no effect of m_1 on y_2 & m_2 on y_1, ∴ there's no process interact' ~~so pairing can be~~ only direct effect

Pair y_1 w/ u_1 & y_2 w/ ~~y_2~~ u_2 since ~~they have~~ these combos have ~~las~~ largest ~~in~~ relative gain





$\boxed{\text{1-1 / 2-2 configuration}}$

$$y_1 = G_{p11}u_1 + G_{p12}u_2 \quad (1), \quad y_2 = G_{p21}u_1 + G_{p22}u_2 \quad (2)$$

if y_{2sp} no change, $y_{2sp} = 0$. $\therefore E_2 = -y_2$.

$\cancel{u_2 = G_c}$ $\quad u_2 = G_{c2}\cdot(E_2) = -G_{c2}y_2 \quad (3)$.

$$(2) \Rightarrow \quad y_2 = G_{p21}u_1 - G_{c2}y_2G_{p22}$$

$$y_2 = (G_{p21}u_1) / 1 + G_{c2}G_{p22}$$

$$(3) \Rightarrow \therefore u_2 = -G_{c2}\frac{G_{p21}}{1 + G_{c2}G_{p22}}\cdot u_1.$$

$$y_1 = G_{p11}u_1 - \frac{G_{c2}G_{p21}}{1 + G_{c2}G_{p22}}\cdot G_{p12}$$

$$y_1 = \left[G_{p11} - \frac{G_{p12}G_{p21}G_{c2}}{1 + G_{c2}G_{p22}}\right]u_1$$

$\cancel{y_1 = G_p}$ $\quad y_{1} = G_{1eff}\,u_1$. $\quad$ effective gain b/w y_1 & u_1 w/ loop 1 & 2 closed.

~~at ss, $G_{p11} = K_{11}$, $G_1 \dots G_{c2}$.~~

$$G_{1eff} \text{ at ss, } K_{11eff} = K_{11} - \frac{(K_{12}K_{21})(G_{c2})}{\left(\frac{1}{G_{c2}} + G_{p22}\right)(G_{c2})}$$

$G_{c2} \to \infty$, $\frac{1}{G_{c2}} \to 0$.

$$K_{11eff} = K_{11} - \frac{K_{12}K_{21}}{K_{22}}$$

$G_c = K_c\left(1 + \frac{1}{\tau s}\right)$

as $s \to 0$

$G_c = \infty$

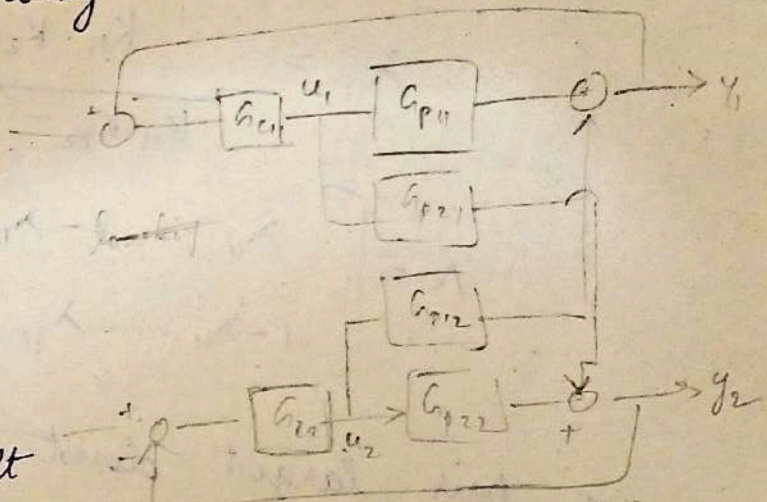
20 a) What is relative gain array? Explain how RGA can be used for loop pairing in a multivariate system. (10)

RGA

- used for quantitative measure of process interactions

$$\frac{\partial y_i}{\partial u_j}$$

- Process interaction: undesirable interaction b/w control loops.
  (eg: Δu_1 changes y_1 from y_{sp1} so u_1 brings it to SS, this cause Δ in y_2 which causes u_2 to try to bring y_2 to y_{sp2}. this effects y_1, etc.
  Brings sys to new steady state.

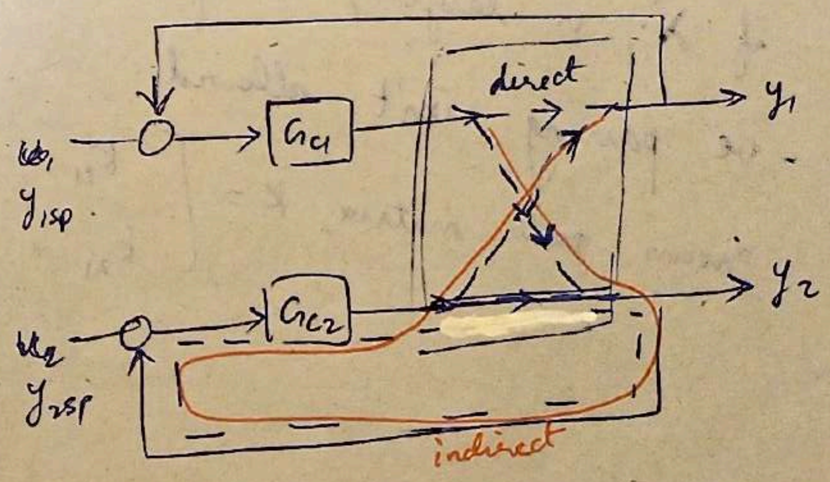


- change to u_1 causes
  - direct effect on y_1
  - indirect effect on y_2 via control loop interactions

- control loop interactions make controller tuning more difficult destabilizes system.

relative gain

$$\lambda_{ij} = \frac{\left(\frac{\partial y_i}{\partial u_j}\right)_u}{\left(\frac{\partial y_i}{\partial u_j}\right)_y}$$



$$= \frac{\text{open loop gain}}{\text{closed loop gain}}$$

ie ~~rel~~ rel. gain is partial derivative of o/p wrt i/p.

$$\Lambda = \begin{bmatrix} \lambda_{11} & \lambda_{12} & \cdots & \lambda_{1n} \\ \lambda_{21} & \lambda_{22} & \cdots & \lambda_{2n} \\ \vdots & & & \vdots \\ \lambda_{n1} & \lambda_{n2} & \cdots & \lambda_{nn} \end{bmatrix}$$

→ dimensionless
- normalised
- measure based only on steady state conditions
- square matrix

$$\boxed{\Lambda = K \otimes (K^{-1})^T}$$

* Λ is element by element multiplication of steady state gain matrix & transpose of its inverse.

$$\lambda_{11} = \frac{K_{11} K_{22}}{K_{11} K_{22} - K_{12} K_{21}} = \lambda_{22}$$

$$\therefore \Lambda = \begin{bmatrix} \lambda_{11} & 1-\lambda_{11} \\ 1-\lambda_{11} & \lambda_{11} \end{bmatrix}$$

* we pair largest closest number to 1 ie.
  if λ_{ij} is largest, we pair c_j w/ y_i.
* -ve pairing isn't allowed.

process gain matrix, $K = \begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix}$.

20 Calculate relative gain of a 2x2 process w/ ss model: M5

$$y_1 = K_{11} u_1 + K_{12} u_2$$
$$y_2 = K_{21} u_1 + K_{22} u_2$$
; or compute relative gain array.

10+6.

relative gain, $\lambda_{ij} = \left(\frac{\partial y_i}{\partial u_j}\right)_u \div \left(\frac{\partial y_i}{\partial u_j}\right)_y$

$$= \frac{\text{OL gain}}{\text{CL gain}}.$$

$$\therefore \lambda_{11} = \left(\frac{\partial y_1}{\partial u_1}\right)_u \div \left(\frac{\partial y_1}{\partial u_1}\right)_y = \frac{1}{1 - \frac{K_{12}K_{21}}{K_{11}K_{22}}}$$

$$\lambda_{11} = \frac{1}{1 - \frac{K_{12}K_2}{K_{11}K_{22}}}.$$

$$\Lambda = K \otimes (K^{-1})^T$$

$$K^{-1} = \frac{\text{adj}(K)}{\det(K)} = \frac{\text{adj}\begin{pmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{pmatrix}}{K_{11}K_{22} - K_{21}K_{12}} = \frac{\begin{bmatrix} K_{22} & -K_{12} \\ -K_{21} & K_{11} \end{bmatrix}}{K_{11}K_{22} - K_{21}K_{12}}$$

$$(K^{-1})^T = \begin{pmatrix} K_{22} & -K_{21} \\ -K_{12} & K_{11} \end{pmatrix} \div (K_{11}K_{22} - K_{21}K_{12})$$

$$\Lambda = \begin{bmatrix} \frac{K_{11}K_{22}}{K_{11}K_{22} - K_{21}K_{12}} & \frac{-K_{12}K_{21}}{K_{11}K_{22} - K_{21}K_{12}} \\ \frac{-K_{12}K_{21}}{\quad} & \frac{K_{11}K_{22}}{\quad} \end{bmatrix} = \begin{pmatrix} \lambda_{11} & \lambda_{12} \\ \lambda_{21} & \lambda_{22} \end{pmatrix}$$

used to design cross controllers.

$\sum$ row $= \sum$ column $= 1$.

$$\therefore \Lambda = \begin{pmatrix} \lambda_{11} & 1-\lambda_{11} \\ 1-\lambda_{11} & \lambda_{11} \end{pmatrix}$$